PRE-ALGEBRA (8TH) · UNIT 3: LINEAR AND FUNCTIONAL RELATIONSHIPS

Road Trip Rate Lab

Compare cars on a road trip using tables, graphs, and equations.

CP2

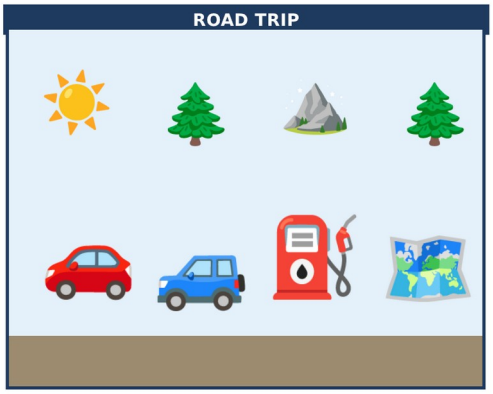
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Your class is planning a road trip to a national park, and three family cars are going. Each car's speed is given a different way: one with a table, one with a graph, and one with an equation.

You will compare the cars, write functions for distance and fuel, and plan when everyone will arrive and how much the gas will cost.



Car A table

0 h → 0 mi · 1 h → 55 mi

2 h → 110 mi · 3 h → 165 mi

Car B graph

passes (1, 80) and (3, 200)

Car C

d = 62t

Learning Goals

- I can find a rate of change (slope) from a table, a graph, or an equation.
- I can write and use a linear function y = mx + b.
- I can tell whether a relationship is linear.

Project Timeline

When	What you will do
Day 1	Car A from a Table
Day 2	Comparing Cars
Day 3	Gas and Patterns
Final	Road Trip Plan + reflection

Part 1: Car A from a Table

Day 1 · Car A: 0 h → 0 miles, 1 h → 55 miles, 2 h → 110 miles, 3 h → 165 miles.

1. What is Car A's rate of change (speed)?

   Starter: *Change in miles ÷ change in hours = ___*

2. Write an equation for Car A's distance d after t hours.

3. How far does Car A go in 4.5 hours?

Part 2: Comparing Cars

Day 2 · Car B starts 20 miles ahead and drives 60 mph: d = 60t + 20. Car C: d = 62t.

4. Which car is faster, Car A or Car C? How do you know?

5. How far from the start is Car B after 3 hours?

6. What do the 60 and the 20 mean in d = 60t + 20?

Part 3: Gas and Patterns

Day 3 · Each car has a 14-gallon tank and gets 30 miles per gallon.

7. Gas left after d miles is $g = 14 - \frac{d}{30}$. How much gas is left after 150 miles?

8. How many miles can a car go before the tank is empty?

9. A square tile with side s has area A: s = 1, 2, 3 → A = 1, 4, 9. Is this relationship linear? Explain.

Final Product: Road Trip Plan

Your final product must include:

- Rates for Cars A, B, and C
- An equation for each car
- Distance after different times
- Gas-remaining function and range of the tank
- A trip poster that compares the cars

Reflection

R1. How can you find a rate of change from a table?

R2. What does the y-intercept mean on a distance graph?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Rates, functions, and comparisons are correct and interpreted in context. Goes beyond the minimum with a creative, realistic choice.	Rates, functions, and comparisons are correct and interpreted in context.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Trip plan shows each car's function, arrival time, and cost with clear labels. Polished, organized, and visually engaging.	Trip plan shows each car's function, arrival time, and cost with clear labels. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.